Adnan Sarhan

AI Engineer

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PROFILE

AI Engineer and Software Engineering graduate specializing in production-grade Generative AI solutions. Experienced in building LLM-powered applications using RAG, agentic AI, tool calling, prompt engineering, and AWS Bedrock/OpenAI. Skilled in designing scalable cloud-native AI systems on AWS with MLOps, CI/CD, and production-ready deployment practices. Passionate about building intelligent AI solutions that solve real-world business challenges.

HIGHLIGHTED SKILLS

- **AI / Generative AI:** LLM Integration (AWS Bedrock), Retrieval-Augmented Generation (RAG), Prompt Engineering, Vector Databases & Embeddings, Agentic Workflows & Tool-Calling.
- **Machine Learning & Data:** Python, SQL, Pandas, NumPy, Scikit-learn, Exploratory Data Analysis (EDA), Feature Engineering, Model Building & Evaluation, Recommendation Systems, NLP.
- **Statistical Analysis:** A/B Testing, Hypothesis Testing (Linear & Non-linear).
- **MLOps & Cloud:** AWS (Bedrock, Lambda, S3), MLOps Pipelines & Deployment, CI/CD, Model Evaluation & Monitoring, Docker, Kubernetes.
- **Engineering & Development:** Flask, JavaScript/TypeScript (Node.js), Java (Spring), React.js, PostgreSQL, MySQL, DynamoDB, Firebase, MongoDB, Agile, Git/GitHub/GitLab, Automated Testing.
- **Languages:** Arabic (Native), English (Fluent).

AI ENGINEERING EXPERIENCE

Data Science Bootcamp - AI/ML, General Assembly, Bahrain **Jun 2026 – Sep 2026**

Completed an intensive, project-based Data Science program spanning the applied ML lifecycle — data wrangling, EDA, feature engineering, model building/evaluation, and deployment — using Python, Pandas, NumPy, and scikit-learn.

AI Software Architect & Developer, AWS CIC Program, Bahrain **Oct 2024 – Jan 2025**

- Designed and developed QuestAI, a cloud-based GenAI platform with a Retrieval-Augmented Generation (RAG) pipeline for automated, standards-aligned exam creation, serving the Ministry of Education.
- Built a responsive React-based frontend and defined infrastructure-as-code using the Serverless Stack (SST).
- Integrated AWS Bedrock for LLM inference alongside Amplify and Cognito (auth), API Gateway and Lambda (backend), DynamoDB (database), S3 (storage), Polly (text-to-speech), Transcribe (speech-to-text), and SNS (notifications) — an end-to-end GenAI application architecture.

OTHER EXPERIENCE

IT Infrastructure and Networking, The Benefit Company, Seef, Bahrain **Jul 2024 – Aug 2024**

- Gained hands-on experience with enterprise-grade physical firewalls and network switches, contributing to the security and reliability of systems serving over 1.3 million users.
- Worked with SolarWinds for proactive network monitoring, performance optimization, and troubleshooting to ensure seamless operations.

- Acquired in-depth knowledge of Cisco Identity Services Engine (ISE), including its role in network access control (NAC) and securing critical resources.

PROJECTS

1. **Lahjaty — Bahraini Arabic Text-to-Speech Model** *(Python, PyTorch, Audio ML, Speech Synthesis*)**:** Developed a zero-shot Bahraini Arabic text-to-speech model by training on 200+ hours of Bahraini podcast audio. Fine-tuned the model through iterative dataset preparation, audio processing, and training experiments until it produced natural-sounding Bahraini speech from unseen voices and text **(Deployed link**: https://bahrain-tts.duckdns.org **)** .

2. **Mashroo3i — AI Analytics Dashboard** *(Python, Machine Learning, AI Agents, Data Visualization)***:** Built an end-to-end analytics dashboard and AI platform for Brinc, combining data visualization with intelligent analytics features. Developed a classification model achieving 0.90 recall and an AI agent that retrieves relevant sources and applies a structured scoring system to generate concrete, data-driven insights.

3. **QuestAI** *(SST, AWS Bedrock, AWS Lambda, AWS DynamoDB, AWS Transcribe, AWS Polly):* A cloud-based, LLM-powered platform with a RAG pipeline to assist the Ministry of Education in creating exams aligned with national competency standards.

4. **StudyAI** *(React.js, Flask, AWS S3, AWS DynamoDB):* An AI-powered platform tailored for University of Bahrain students, providing LLM-generated, customized study aids and resources.

EDUCATION

- **BSc. Software Engineering,** University of Bahrain, Zallaq, Bahrain **Sep 2020 – Jan 2025**

CERTIFICATION

- **AWS Certified Cloud Practitioner,** Amazon Web Services (AWS), Bahrain **2024**
- **CCST Cybersecurity,** Cisco, Bahrain **2026**